# GURU GOBIND SINGH INDRAPRASTHA UNIVERSITY

# PRACTICAL FILE

## B.TECH(CSE)

## Programming in Java (CIC - 258)
## 2025-2026

**SUBMITTED TO**

**Ms. Garima Maharaj Singh**
**Assistant Professor**

**SUBMITTED BY**

**Name :**
**Enrolment No :**
**Class : 2nd Year**
**Section :**

**JIMS ENGINEERING MANAGEMENT TECHNICAL CAMPUS**

**48/4, KNOWLEDGE PARK, GREATER NOIDA, UP-201306**

**(AFFILIATED TO GGSIPU)**

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## PROGRAM 1

**Q1. Write a program to sort the elements of an array in descending order.**

```
import java.util.Scanner;
public class DescendingSort {
   public static void main(String[] args) {
       Scanner sc = new Scanner(System.in);
      System.out.println("Name: Samakhya Sharma");
       System.out.println("Q1. Program to Sort Array in Descending Order");
       System.out.println("--------------------------------------------");
      System.out.print("Enter number of elements: ");
       int n = sc.nextInt();
       int arr[] = new int[n];
      System.out.println("Enter elements:");
       for(int i = 0; i < n; i++) {
           arr[i] = sc.nextInt();
       }
      for(int i = 0; i < n - 1; i++) {
           for(int j = 0; j < n - i - 1; j++) {
               if(arr[j] < arr[j + 1]) {   // Descending condition
                   int temp = arr[j];
                   arr[j] = arr[j + 1];
                   arr[j + 1] = temp;
               }
           }
       }
      System.out.println("Array in Descending Order:");
       for(int i = 0; i < n; i++) {
           System.out.print(arr[i] + " ");
       }
       sc.close();
   }
}
```

**OUTPUT 1:**

```
Name: Samakhya Sharma
Q1. Program to Sort Array in Descending Order
---------------------------------------------
Enter number of elements: 5
Enter elements:
4
8
5
9
1
Array in Descending Order:
9 8 5 4 1
```

**PROGRAM 2**

**Q2. Write a java program to maintain student academic records and display complete details for each student.**

```
import java.util.Scanner;
public class Student {
   int rollNo;
   String name;
   String course;
   int semester;
   float marks1, marks2, marks3;
   float total, percentage;
   void inputDetails() {
      Scanner sc = new Scanner(System.in);
      System.out.print("Enter Roll Number: ");
      rollNo = sc.nextInt();
      sc.nextLine();
      System.out.print("Enter Student Name: ");
      name = sc.nextLine();
      System.out.print("Enter Course Name: ");
      course = sc.nextLine();
      System.out.print("Enter Semester: ");
      semester = sc.nextInt();
      System.out.print("Enter Marks of Subject 1: ");
      marks1 = sc.nextFloat();
      System.out.print("Enter Marks of Subject 2: ");
      marks2 = sc.nextFloat();
      System.out.print("Enter Marks of Subject 3: ");
      marks3 = sc.nextFloat();
      total = marks1 + marks2 + marks3;
      percentage = total / 3; }
   void displayDetails() {
      System.out.println("\n--- Student Academic Record ---");
      System.out.println("Roll Number   : " + rollNo);
      System.out.println("Name          : " + name);
      System.out.println("Course        : " + course);
      System.out.println("Semester      : " + semester);
      System.out.println("Total Marks   : " + total);
      System.out.println("Percentage    : " + percentage + "%");   }
   public static void main(String[] args) {
      System.out.println("Name: Samakhya Sharma");
      System.out.println("Q2. Student Academic Report");
      System.out.println("-----------------------------------");
      Student s = new Student();
```

```
        s.inputDetails();
        s.displayDetails();  }
}
```

**OUTPUT 2:**

```
Name: Samakhya Sharma
Q2. Student Academic Report
-----------------------------------
Enter Roll Number: 20
Enter Student Name: Samakhya
Enter Course Name: Btech CSE
Enter Semester: 4
Enter Marks of Subject 1: 89
Enter Marks of Subject 2: 89
Enter Marks of Subject 3: 92

--- Student Academic Record ---
Roll Number    : 20
Name           : Samakhya
Course         : Btech CSE
Semester       : 4
Total Marks    : 270.0
Percentage     : 90.0%
```

**PROGRAM 3**

**Q3. Write a java program to model an organisation having different categories of employees whose salary is computed differently.**

```
import java.util.Scanner;
public class Employee {
    int empId;
    String name;
    double salary;
    void inputDetails() {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Employee ID: ");
        empId = sc.nextInt();
        sc.nextLine();
        System.out.print("Enter Employee Name: ");
        name = sc.nextLine();
    }
    void calculateSalary() {
   }
    void displayDetails() {
        System.out.println("\nEmployee ID    : " + empId);
        System.out.println("Employee Name  : " + name);
        System.out.println("Salary         : " + salary);
    }
   public static void main(String[] args) {
        System.out.println("Name: Samakhya Sharma");
        System.out.println("Q3. Categories of Employees");
        System.out.println("----------------------------------");
        Scanner sc = new Scanner(System.in);
        System.out.println("1. Full Time Employee");
        System.out.println("2. Part Time Employee");
        System.out.print("Choose Employee Type: ");
        int choice = sc.nextInt();
        Employee emp;
        if (choice == 1) {
            emp = new FullTimeEmployee();
        } else {
            emp = new PartTimeEmployee();
        }
        emp.inputDetails();
        emp.calculateSalary();
        emp.displayDetails();
    }
}
class FullTimeEmployee extends Employee {
```

```
    double basic, hra, da;

    void calculateSalary() {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Basic Salary: ");
        basic = sc.nextDouble();
        hra = 0.20 * basic;
        da = 0.10 * basic;
        salary = basic + hra + da;
    }
}
class PartTimeEmployee extends Employee {
    int hoursWorked;
    double ratePerHour;
    void calculateSalary() {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Hours Worked: ");
        hoursWorked = sc.nextInt();
        System.out.print("Enter Rate per Hour: ");
        ratePerHour = sc.nextDouble();
        salary = hoursWorked * ratePerHour;
    }
}
```

**OUTPUT 3:**

```
Name: Samakhya Sharma
Q3. Categories of Employees
-----------------------------------
1. Full Time Employee
2. Part Time Employee
Choose Employee Type: 1
Enter Employee ID: 20
Enter Employee Name: Samakhya
Enter Basic Salary: 15000

Employee ID     : 20
Employee Name   : Samakhya
Salary          : 19500.0
```

**PROGRAM 4**

**Q4. Develop a Java program to implement a digital payment system supporting multiple payment modes with different processing logic.**

```
import java.util.Scanner;
public abstract class Payment {
    double amount;
    abstract void processPayment();
    void inputAmount() {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Payment Amount: ");
        amount = sc.nextDouble();
    }
   public static void main(String[] args) {
        System.out.println("Name: Samakhya Sharma");
        System.out.println("Q4. Digital Payment System");
        System.out.println("-----------------------------------");
        Scanner sc = new Scanner(System.in);
        System.out.println("1. Credit Card Payment");
        System.out.println("2. UPI Payment");
        System.out.println("3. Cash Payment");
        System.out.print("Select Payment Mode: ");
        int choice = sc.nextInt();
        Payment payment;
        switch (choice) {
            case 1:
                payment = new CreditCardPayment();
                break;
            case 2:
                payment = new UPIPayment();
                break;
            case 3:
                payment = new CashPayment();
                break;
            default:
                System.out.println("Invalid Choice");
                return;
        }
        payment.inputAmount();
        payment.processPayment();
    }
}
class CreditCardPayment extends Payment {
    void processPayment() {
        System.out.println("Processing Credit Card Payment...");
```

```
        System.out.println("Payment of Rs. " + amount + " completed using Credit
Card.");
    }
}
class UPIPayment extends Payment {
    void processPayment() {
        System.out.println("Processing UPI Payment...");
        System.out.println("Payment of Rs. " + amount + " completed using UPI.");
    }
}
class CashPayment extends Payment {
    void processPayment() {
        System.out.println("Processing Cash Payment...");
        System.out.println("Payment of Rs. " + amount + " received in cash.");
    }
}
```

**OUTPUT 4:**

```
Name: Samakhya Sharma
Q4. Digital Payment System
-----------------------------------
1. Credit Card Payment
2. UPI Payment
3. Cash Payment
Select Payment Mode: 2
Enter Payment Amount: 500
Processing UPI Payment...
Payment of Rs. 500.0 completed using UPI.
```

## PROGRAM 5

**Q5. Create a customised exception and also make use of all the 5 exception keywords.**

```
import java.util.Scanner;
public class InvalidAgeException extends Exception {
    InvalidAgeException(String message) {
        super(message);
    }
   static void checkAge(int age) throws InvalidAgeException {
        if (age < 18) {
            throw new InvalidAgeException("Age must be 18 or above!");
        } else {
            System.out.println("Eligible to vote.");
        }
    }
   public static void main(String[] args) {
        System.out.println("Name: Samakhya Sharma");
        System.out.println("Q5. Custom Exception using 5 Exception Keywords");
        System.out.println("-----------------------------------");
        Scanner sc = new Scanner(System.in);
        try {
            System.out.print("Enter your age: ");
            int age = sc.nextInt();
            checkAge(age);
        } catch (InvalidAgeException e) {
            System.out.println("Exception Caught: " + e.getMessage());
        } finally {
            System.out.println("Program executed successfully.");
            sc.close();
        }
    }
}
```

**OUTPUT 5:**

```
Name: Samakhya Sharma
Q5. Custom Exception using 5 Exception Keywords
-----------------------------------
Enter your age: -12
Exception Caught: Age must be 18 or above!
Program executed successfully.
```

```
Name: Samakhya Sharma
Q5. Custom Exception using 5 Exception Keywords
-----------------------------------
Enter your age: 19
Eligible to vote.
Program executed successfully.
```